INCH-POUND
MS45904C
07 July 2022
SUPERSEDING
MS45904B
06 February 1974

DETAIL SPECIFICATION SHEET

WASHER, LOCK, INTERNAL AND EXTERNAL TOOTH

Reactivated after 07 July 2022 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the described herein shall consist of this specification sheet and specification ASME B18.21.1.

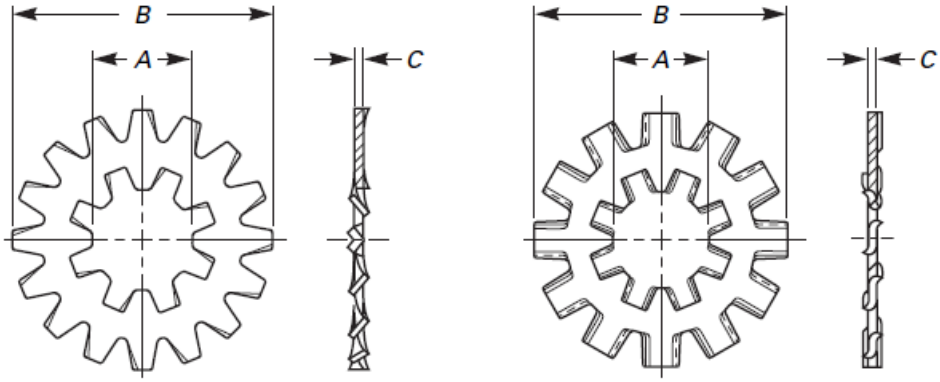


FIGURE 1. Acceptable types of internal and external tooth lock washer.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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REQUIREMENTS:

1 Steel composition. Unless otherwise specified, washers shall be made from carbon steel, SAE 1050 through 1065 (UNS G10500–G10650), (see SAE J403, Table 1).

2 Hardness. Washers shall be hardened to HRC 40 through HRC 50 or equivalent measuring method.

3 Internal and external teeth. Internal and external teeth shall be either Type A or Type B in accordance with ASME B18.21.1.

4 Protective finish. Zinc-Nickel plate in accordance with MIL-PRF-32647, Type K20, or Cadmium[1/] in accordance SAE AMS-QQ-P-416, Type II, Class 2.

[1/] Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647 is the preferred method.

5 Conformance. Certificates of compliance to paragraph 1, 2, and 3 shall be provided to verify conformance.

6 Part or Identifying Number (PIN). The PIN consists of the following:

MS45904 – 818

- 818: Dash number from Table I indicating size/dimensions (cross-referenced to Figure 1.)
- MS45904: Specification sheet number

Example: MS45904-818 is a nominal 0.250 inch internal and external tooth lock washer, outside diameter (OD) approximately 0.750 inch with Zinc-Nickel plating.

TABLE I. Dash numbers and dimensions (inch).

DASH NUMBER ZnNi	NOMINAL SIZE	INSIDE DIAMETER A		OUTSIDE DIAMETER B		THICKNESS C	
		MAX	MIN	MAX	MIN	MAX	MIN
801	0.112	0.123	0.115	0.475	0.460	0.021	0.016
804	0.138	0.150	0.141	0.510	0.495	0.028	0.023
807	0.164	0.176	0.168	0.610	0.580	0.034	0.028
808	0.164	0.176	0.168	0.690	0.670	0.034	0.028
809	0.164	0.176	0.168	0.760	0.740	0.034	0.028
810	0.190	0.204	0.195	0.610	0.580	0.036	0.028
811	0.190	0.204	0.195	0.690	0.670	0.040	0.032
812	0.190	0.204	0.195	0.760	0.740	0.040	0.032
813	0.190	0.204	0.195	0.900	0.880	0.040	0.032
818	0.250	0.267	0.256	0.760	0.725	0.040	0.032
819	0.250	0.267	0.256	0.900	0.880	0.040	0.032
820	0.250	0.267	0.256	0.985	0.965	0.045	0.037
821	0.250	0.267	0.256	1.070	1.045	0.045	0.037
822	0.3125	0.332	0.320	0.900	0.865	0.040	0.032
823	0.3125	0.332	0.320	0.985	0.965	0.045	0.037
824	0.3125	0.332	0.320	1.070	1.045	0.050	0.042
826	0.375	0.398	0.384	0.985	0.965	0.045	0.037
827	0.375	0.398	0.384	1.070	1.045	0.050	0.042
828	0.375	0.398	0.384	1.155	1.130	0.050	0.042
829	0.375	0.398	0.384	1.260	1.220	0.050	0.042
834	0.500	0.530	0.512	1.260	1.220	0.055	0.047
836	0.500	0.530	0.512	1.410	1.380	0.060	0.052
837	0.500	0.530	0.512	1.620	1.590	0.067	0.059
842	0.625	0.663	0.640	1.410	1.380	0.060	0.052
843	0.625	0.663	0.640	1.620	1.590	0.067	0.059

NOTE: Washers cannot be used when directly converted to metric, as such, no metric conversion is given.

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TABLE II. Inactive and replacement part numbers.

PIN				
Inactive for new Design	**Zinc-Nickel Replacement**		**Inactive for new Design**	**Zinc-Nickel Replacement**
MS45904 -	MS45904 -		MS45904 -	MS45904 -
51	801		74	824
52	*		75	*
53	*		76	826
54	804		77	827
55	*		78	828
56	*		79	829
57	807		80	*
58	808		81	*
59	809		82	*
60	810		83	*
61	811		84	834
62	812		85	*
63	813		86	836
64	*		87	837
65	*		88	*
66	*		89	*
67	*		90	*
68	818		91	*
69	819		92	842
70	820		93	843
71	821		94	*
72	822		95	*
73	823			

* Denotes no replacement part number, previously designated as inactive for new design.

7 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids. Copies of MIL-PRF-32647 are available online at https://quicksearch.dla.mil. Copies of ASME B18.21.1 are available online at https://www.asme.org. Copies of SAE J403 and SAE AMS-QQ-P-416 are available online at https://www.sae.org.

8 Design. For design feature purposes, this document takes precedence over all other documents.

9 Dimensions. Unless otherwise specified, all dimensions are in inches.

10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 99
DLA – GS

Review activities:
Army – AR

Preparing Activity:
Army – AT

(Project 5310-2022-026)

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